Jiwoo Chung

Software Developer Intern

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Education

George Mason University — Fairfax, VA
Bachelor of Science in Computer Science — ***Expected Dec 2027***

Relevant Coursework: Software Engineering, Data Structures and Algorithms, Computer Systems, Linear Algebra
Certificate: IBM Machine Learning (2024)
Leadership/ Activities: GMU LeetCode Club Treasurer, **GMU Korean Student Association** Fundraising Chair, **MoviesML** Project Manager, U.S. National Taekwondo Sparring Team Member

Skills

Technical: Java, Python, C, SQL, JavaScript, HTML/CSS, Pine Script, React, Spring Boot, Node.js, NumPy, SciPy, Scikit-learn, MNE, YOLO, MobileSAM, PostgreSQL, Docker, Git, Linux, Jira, Docker
AI Development Tools: OpenAI Codex, Claude, LHXGPT
Technologies & Concepts: REST APIs, JFrog Artifactory, Machine Learning, Computer Vision, Signal Processing, Microservices, Distributed Systems, Unit Testing, Agile, Containerization

Work Experience

L3Harris Technologies | Space and Mission — Chantilly, VA
Software Engineering Intern — *May 2026 – Aug 2026*

- Engineered a scalable full-stack constellation modeling application using **React**, **Spring Boot**, and **PostgreSQL**, designing **10 REST API endpoints** to configure constellations, ground stations and mission simulation scenarios.
- Containerized **4 application services** with **Docker Compose**, integrating container images through **Artifactory** in **Linux** environments, reducing local setup time by **1+ hour per developer** across an **11-person team**.
- Resolved a blocked **AGI/Ansys STK Engine** integration under tight deadlines, enabling a functional MVP and live demonstration for **5+ executive leaders**, including the Vice President.

College of William & Mary | OceanicAI — Williamsburg, VA
AI Engineering Intern — *Jan 2026 – May 2026*

- Engineered an end-to-end computer vision pipeline using **YOLOv11 Nano** and **MobileSAM** to detect, segment, and estimate the millimeter-scale size of sea scallops across **3,600 survey images**.
- Preprocessed **10 years** of Mid-Atlantic marine survey data by separating double-frame imagery into **640×640** images and fine-tuning **COCO-pretrained** model weights for downstream object detection and segmentation.
- Optimized model training across **50 epochs** on a **T4 GPU** to support efficient multi-year image analysis; the project earned **Best Project Runner-Up** and a **$1,000 prize** among **12 teams**.

George Mason University — Fairfax, VA
Intro to CS Undergraduate Teaching Assistant — *Aug 2024 – May 2025*

- Supported **500+ students** with Java programming, object-oriented programming, data structures, and debugging, translating technical concepts into actionable guidance and diagnosing implementation errors.

Projects

BCI EEG Motor Imagery Classification | *Python, NumPy, SciPy, Scikit-learn, MNE* — GitHub

- Engineered an EEG signal-processing pipeline in **Python** to transform **68-channel, 512 Hz** neural recordings into **200 labeled motor-imagery trials** for left- and right-hand command classification.
- Applied **8–30 Hz Butterworth filtering**, log-variance feature extraction, and motor-cortex channel analysis to isolate discriminative Mu/Beta activity from raw EEG signals.
- Increased classification accuracy from **40% to 57.5%** by integrating **Common Spatial Patterns (CSP)** with **LDA** and testing sliding-window augmentation across subjects.

BoS and FVG Algorithmic Trading Strategy | *Pine Script, TradingView*

- Engineered a multi-timeframe **Pine Script** strategy to detect **Break of Structure** and **Fair Value Gaps**, tracking swing levels, gap boundaries, and event sequencing across **15-minute, 1-hour, and 4-hour** market data.
- Automated **entry, exit, take-profit, and stop-loss** execution, achieving a **70%+ backtested win rate** with a target risk-reward ratio above **3:1**.